L-C-1021C
April 30, 1979
SUPERSEDING
L-C-001021B,
April 25, 1972 and
L-C-001081B, April 25, 1972

FEDERAL SPECIFICATION

COVER, MATTRESS, AND PILLOW (PLASTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for a plastic mattress and pillow cover.

1.2 Classification.

1.2.1 Type and size. The covers shall be of the following types and sizes (see 6.2).

Type I mattress cover.

Size 1 - 36 inches wide by 76 inches long by 6-1/2 inches high.
Size 2 - 36 inches wide by 81 inches long by 6-1/2 inches high.

Type II pillow cover.

Size 3 - 20 inches wide by 27 inches long.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-F-106 - Fasteners, Slide, Interlocking.
PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 882 - Tensile Properties of Thin Plastic Sheetings.
D 1433 - Flammability of Flexible Thin Plastic Sheeting.
D 1790 - Brittleness Temperature of Plastic Film by Impact.
D 1922 - Propagation Tear Resistance of Plastic Film and Thin Sheeting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Standard Test Methods for Vinyl and Pyroxylin Coated Fabrics and Vinyl Sheeting.

(Application for copies should be addressed to the Vinyl Fabrics Institute, 60 East 42nd Street, New York, NY 10017.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Vinyl film. The material shall be vinyl film, 8 gage (0.008 ± 10 %) in thickness for the mattress cover and 4 gage (0.004 ± 10 %) in thickness for the pillow cover before embossing. The vinyl film shall be boilable and contain no ingredients known to be skin sensitizers or irritants and shall conform to the physical requirements as shown in table I, when tested as specified in 4.5. Test in table III shall be conducted on the vinyl film after embossing unless otherwise indicated. The vinyl film shall meet the resistance requirements specified in 3.1.1 and 3.1.2. The material shall be what is commercially known as taffeta embossed on one side.

TABLE I. Physical requirements

Characteristic	Requirements	
	Pillow cover	Mattress cover
Tensile strength, pounds/sq inch, minimum	2,000	2,300
Elongation, percent, minimum	180	180
Tear strength, grams/mil, minimum (machine and cross direction	60	90
Dimensional stability, percent change, maximum (plus or minus)	10.0	10.0
Volatility, percent loss, maximum	4.0	2.1
Rate of burn inches/second, maximum	1.2	1.2
Low temperature impact at 0°F for mattress covers and 15°F for pillow covers	pass	pass

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3.1.1 Resistance to boiling water. The vinyl film shall not become brittle or sticky, when tested as specified in 4.5.2.

3.1.2 Stain resistance. The vinyl film shall show no effects, other than minimum discoloration, to blood, saliva, grease, or perspiration, when tested as specified in 4.5.4.

3.1.3 Slide fastener. The slide fastener for the closure shall be type I, size LM, style 6, short pull tab, aluminum alloy conforming to Federal Specification V-F-106. The tape for the slide fastener shall be of the material specified in 3.1 or it shall be a white cotton herringbone twill. The tape shall be of sufficient width and strength for efficient operation of the fastener and secure attachment to the cover.

3.1.4 Eyelet (mattress cover). The metal eyelet, to be used as an air vent in the mattress cover, shall be circular of a noncorrosive metal and shall have a hole 1/4 ± 1/16 inch in diameter.

3.1.5 Thread and stitching (pillow cover). The thread for sewing the slide fastener and seaming the pillow cover shall be natural color nylon. The seam shall be a bounding type using a binding strip and the stitch type shall be interlacing loops of 2 thread groups. The stitching shall be not less than 1/4 inch from the edge of the binding with not less than 6 nor more than 8 stitches per inch.

3.2 Design and construction. Design and construction for the mattress and pillow covers are as follows:

3.2.1 Mattress cover construction. The mattress cover shall be of a two-piece construction (body construction is one piece and panel for slide fastener is one piece) cut from the material specified in 3.1. The cover shall be electronically heat sealed and formed to provide a cover with a box type effect to conform to measurements specified in 3.3.1 with slide fastener closure at one end.

3.2.1.1 Detailed construction for mattress cover. A single piece of vinyl film shall be folded end to end to form a double thickness and shall be of sufficient size to meet finished dimensions as specified in 3.3.1. The raw edges of the cover sides shall be seamed to within 8 inches on each side from open end. The corners at the closed end shall be constructed and seamed to provide for the required boxing effect and height (3.3.1) all around the cover. All seams shall be electronically heat sealed 1/2 inch from raw edges. The material in the 8 inch area on each side at the open end shall be taper cut or otherwise constructed for secure attachment of the slide fastener panel and to provide firm corners. The slide fastener panel shall be cut to fit across the open end and 8 inches along both sides and retaining the box effect. Each end of the panel shall be taper cut or otherwise constructed for secure attachment to the cover. The panel shall be center cut or slit lengthwise and the slide fastener electronically heat sealed in the panel. The slide fastener shall extend midway along the entire front of the boxed area and 6 inches down each side. The slide fastener for the closure shall be not less than 47 inches long. The slide fastener panel shall be electronically heat sealed to the top, bottom, and along the sides of the cover. There shall be no raw edges on the outside of the cover or open seams remaining after the seaming operations. A tab, 1 inch wide by 2 inches long, shall be attached to each side wall to pull the cover over the mattress. A minimum of one eyelet 1/4 inch in diameter shall be attached and centered in each side. See Figure 1 which covers 36 inches wide by 76 inches long by 6-1/2 inches high mattress cover.

3.2.2 Pillow cover construction. The pillow cover shall be fabricated from one panel of the material specified in 3.1. The vinyl film shall be folded to form a flat bag. The pillow cover shall be bound on two sides and open end with the material specified for the pillow cover in 3.1. The binding tape for the sides and open end shall be 1 (+ 1/8) inches wide and shall have a 3/16 (± 1/16) inch turn-under. One row of stitching shall be used to join the two plies of material and binding together. The slide fastener shall be not less than 19 inches long and shall have a stop on each end. The fastener shall be centered crosswise and shall be securely attached to one end (open end) with binding tape. The slide fastener tape shall be extended 3/4 inches on each end, past the end of the fastener chain. Edges of the slide fastener tape shall be sewed on ends and sides, in order to remain fast and secure under normal opening and closing of the fastener.

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3.2.3 Seams. Seams shall be continuous and shall not weaken the basic material. When tested as specified in 4.5.1, the strength of the bonded seams shall be not less than 65 percent of the breaking strength of the film.

3.3 Size.

3.3.1 Dimensions and tolerances. The finished minimum dimensions of the mattress cover shall be 36 inches wide by 76 inches long by 6-1/2 inches high or 36 inches wide by 81 inches long by 6-1/2 inches high, as specified (see 6.2). The following tolerances will be permitted in the dimensions: Length - minus 0, plus 1-1/2 inches. Width - minus 0, plus 1 inch. Height - minus 0, plus 1/2 inch. The pillow cover shall be 20 inches wide by 27 inches long with the following tolerances; width ± 1/4 inch; length ± 1/4 inch.

3.4 Color and finish. The mattress and pillow cover shall be white and the taffeta embossed side of the vinyl film shall be on the outside of the cover.

3.5 Workmanship. The finished covers shall be free from defects which detract from their appearance or may impair their serviceability. The covers shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality level (AQL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Inspection for end item lot acceptance. Sampling for inspection and lot acceptance shall be determined in accordance with MIL-STD-105.

4.3 Visual examination of the end item. Defects found during this examination shall be classified in accordance with table II. The lot size shall be expressed in terms of one cover. The unit of product shall be expressed in the same terms as the lot size. The applicable inspection level for the cover shall be level S-4 and the AQL shall be 2.5 major defects per hundred units and 10.0 (major and minor combined) defects per hundred units.

4.3.1 Dimensional examination. Any measurement deviating from the specified dimension by more than the tolerance shall be considered a defect. The lot size shall be expressed in units of one cover. The sample unit shall be one cover. The AQL for this examination shall be 2.5 percent defective. The sample size shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking comply with requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Material	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.

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TABLE II. Classification of defects

Examine	Defects	Classification Major	Classification Minor
Vinyl film (mattress and pillow cover)	Bubble, sharp crease, wrinkle, blister, gouge, or other imperfection.		X
	Cut, tear, hole, break, or crack.	X	
	Color not as specified.	X	
	Vinyl film not taffeta embossed on outside of cover.		X
	Objectionable odor (other than odor typical to vinyl chloride).		X
	Any spot, stain or other discoloration affecting appearance.		X
Slide fastener (mattress and pillow cover)	Material other than specified.	X	
	Size not as specified.	X	
	Any part fractured, split or broken.	X	
	Does not operate freely, smoothly or positively, i.e. binds, catches or sticks at any point.		X
	Slider goes beyond stop.	X	
	Slider disengages from either row of scoops when chain is in closed position.	X	
	Slider pull becomes detached during operation.	X	
Eyelet (mattress cover)	Material not metal.	X	
	Inside dimension less than specified.		X
	Eyelet omitted.		X
Construction (pillow cover)	Not a single panel of material.	X	
	Binding tape not of same material as the cover.	X	
Seams and stitching (pillow cover)	Not stitched with interlacing loops of 2 thread groups.	X	
	Less than 6 or more than 8 stitches per inch.		X
	Any open seam.	X	
	Loose seams, tight puckering seams, pleated or bulging edges.		X
Construction of mattress cover	Not of a two piece construction, body one piece and panel for slide fastener, one piece.	X	
	Corners at the closed end not constructed and seamed to provide for boxing effect and height required all around the cover.	X	
	Material in 8 inch approximate area at the open end not taper cut or otherwise constructed.	X	
	Slide fastener panel not center cut or slit lengthwise.	X	
	Tab not of a double thickness of material.	X	
	Tab not in location specified.		X
	One tab omitted.	X	
Bonded seams (mattress cover)	Size not as specified.	X	
	Not completely bonded.	X	
	Puckered or distorted in heat sealing.		X

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4.5 End item testing. The methods of testing specified in table III, and 4.5.1, 4.5.2, 4.5.3, 4.5.4, and 4.5.5, wherever applicable, shall be followed. The physical and chemical values specified in section 3 apply to the determination made on a sample unit for test purposes as specified in the applicable test method as listed in table III. The lot size shall be expressed in units of covers and the test sample unit shall be one cover using inspection level S-2 with an AQL of 4.0 percent defective.

TABLE III. Test methods 1/

Characteristic	Requirement reference	Test method
Tensile strength	Table I 2/	ASTM D 882 method A
Elongation	Table I	ASTM D 882 method A
Tear strength	Table I 2/	ASTM D 1922
Dimensional stability (Dimensional change at elevated temperatures)	Table I	4.5.4
Rate of burn	Table I	ASTM D 1433
Volatility	Table I	4.5.5
Low temperature impact	Table I	ASTM D 1790
Bond strength of heat sealed seams	3.2.3	4.5.2
Resistance to boiling water	3.1.2	4.5.3
Stain resistance	3.1.3	4.5.5

1/ Tests in table III shall be conducted on the vinyl film after embossing, unless otherwise indicated.
2/ Thickness used in calculation shall be the thickness of the vinyl film before embossing.

4.5.1 Bond strength of heat sealed seams. The bond strength of the heat seal shall be tested in the same manner as the tensile strength (table I) except the width of the specimens shall be 1 inch with the seam running in the width direction and located in the center of the specimen. The final results shall be the average of the tests. Specimens shall be prepared as follows:

(a) Two specimens, 1 inch wide by 10 inches long, shall be cut from each corner heat seal.
(b) One specimen, 1 inch wide by 10 inches long, shall be cut from each side, at any point between the corner opening and the closed end of the cover (exclude side with zipper).

4.5.2 Resistance to boiling water. The vinyl film shall be placed in boiling water for 60 minutes and upon removing shall not be sticky or brittle.

4.5.3 Dimensional stability (dimensional change at elevated temperatures). Dimensional stability shall be determined by exposing a 10-inch square sample of film to 212° ± 2°F (100° ± 1.1°C) for 30 minutes ± 1 minute in a circulating air oven. The sample shall be cut from the center of the film sheet with a template accurate to ± 0.02 inches in all directions. The sample shall be placed between sheets of heavy wrapping paper, 15 inches square, lightly dusted with talc. The paper sheets shall be clipped or stapled together, care being taken not to hold or restrict the test sample. The sandwich shall then be placed horizontally in an oven. After the oven exposure, the sample shall be cooled to room temperature and measured to the nearest 0.01 inch along both central axes.

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4.5.4 Stain resistance.

4.5.4.1 Stain resistance to blood. Pour 5 milliliters (ml) of blood (warm-blooded animal or fowl) on a specimen of material 12 inches square. Spread the blood to a thin even layer. After drying, the dried blood shall be allowed to remain in contact with the plastic film for 24 hours. The specimen shall then be washed off with a cloth or sponge using soap and water. Observe for stains or discoloration.

4.5.4.2 Stain resistance to saliva. Use the same method as specified in 4.5.4.1, except that human saliva shall be substituted for the blood.

4.5.4.3 Stain resistance to grease. Pour 5 grams of petrolatum on a specimen of material 12 inches square. Spread the grease to a thin even layer. Allow the grease to remain in contact with the plastic film for 24 hours. Wash off with a cloth or sponge using soap and water. Observe the stains or discoloration.

4.5.4.4 Stain resistance to perspiration. Pour 5 ml of acid solution as specified below, onto a specimen of material 12 inches square. Spread the solution to a thin layer. After drying, allow the residue to remain in contact with the plastic film for 24 hours. Wash off with a cloth or sponge using soap and water. Observe for stains or discoloration. Repeat the testing using alkaline solution as specified below.

Acid solution	Alkaline solution
10 gm sodium chloride	10 gm sodium chloride
1 gm lactic acid U.S.P, 85 percent	4 gm ammonium carbonate U.S.P.
0.25 gm histidine monohydrochloride	0.25 gm histidine monohydrochloride
1 gm disodium orthoposphate, anhydrous	1 gm disodium orthophosphate, anhydrous
If the pH is not 3.5 adjust with lactic acid or disodium orthopo-sphate.	If the pH is not 8.0 adjust with disodium orthoposphate or ammonium carbonate.

4.5.5 Volatility. The test specimen shall be three 2-inch diameter disks which shall be preconditioned at 23° ± 1.1°C (73.5 ± 2°F) and 50 ± 4 percent relative humidity for 48 hours. After preconditioning, the three specimens shall be individually weighed and designated as W_1. One hundred and twenty cubic centimeters of activated carbon shall be evenly spread on the bottom of a container. One specimen shall be placed on top of the activated carbon and this shall be covered with 120 cc of activated carbon. A second specimen shall be placed in the container, covered with 120 cc of carbon, followed by a third specimen, and then 120 cc more of activated carbon. A cover shall be placed on the container in such a manner that the container will be vented. This is necessary in order that possible pressure build-up in the container during heating is relieved. In no case shall the carbon be packed with pressure other than the weight of the composite sandwich in the container. The container shall be placed upright in the oven or bath. The temperature of the oven or bath shall be 70° ± 1°C and the test duration shall be 24 hours. At the end of the 24-hour period, the container shall be removed from the oven and allowed to cool to room temperature, after which time the specimens shall be removed from the container, brushed free of carbon, and reconditioned for 48 hours at 23° ± 1.1°C (73.5 ± 2°F) and 50 ± 4 percent relative humidity. At the termination of the reconditioning period, the specimens shall be reweighed on the analytical balance and the final weight shall be designated as W_2.

Calculations. Percent weight loss. The plasticizer volatility shall be expressed as the percentage weight loss compared to the weight of the original specimen as follows:

$$\text{Percent weight loss} = \frac{W_1 - W_2}{W_2} \times 100$$

where:
W_1 = initial weight of a test specimen
W_2 = final weight of a test specimen

Percent weight loss of the sample. The values obtained for the three specimens shall be averaged and reported as percentage weight loss of the sample tested.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each cover shall be compactly folded, packaged in a bag or wrap and secured.

5.1.2 Commercial. The covers shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Six covers packaged as specified in 5.1; shall be packed in a box conforming to PPP-B-636, grades V3c, V3s, or V2s. The boxes shall be closed, waterproofed, and reinforced in accordance with the appendix to PPP-B-636. Alternatively wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive.

5.2.2 Commercial. The covers packaged as specified in 5.1, shall be packed in fiberboard boxes that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification.

5.2.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized to facilitate handling, in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The mattress cover covered by this specification is intended primarily for use in hospitals for the purpose of protecting mattresses and pillows from soilage due to excretions or secretions.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2.1).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.3).

6.3 Safety hazards for hospital use. Because of possible electrostatic buildup, the plastic covers covered by this specification should not be used when oxygen is administered by the tent method or where flammable anesthetics are used (National Fire Protection Association Bulletin No. 56 - Code for the Use of Flammable Anesthetics (current edition). Local safety regulations should be followed regarding materials to be used under these conditions.

MILITARY INTEREST:

DOD has waived coordination on revisions and amendments to this federal specification until further notice.

Preparing activity:

GSA-FSS

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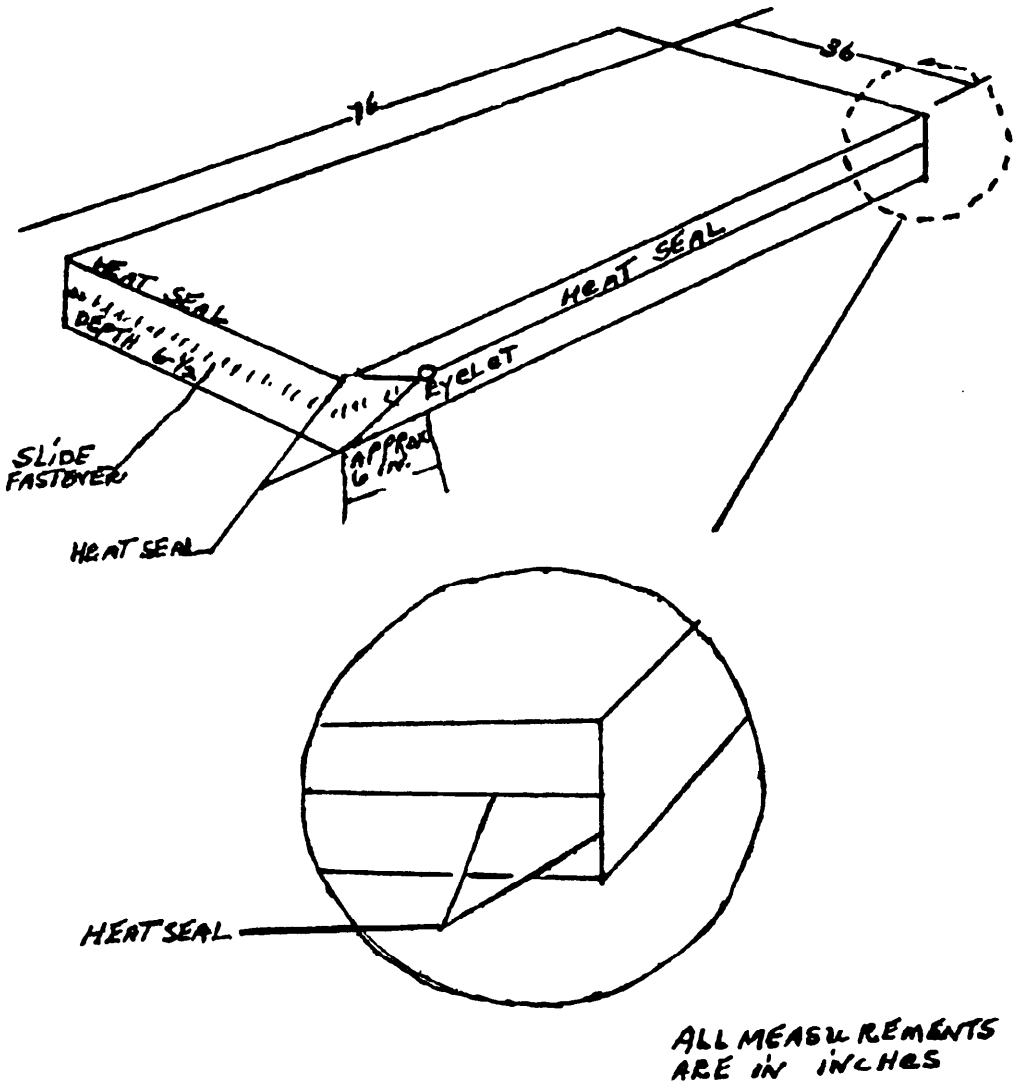


FIGURE 1

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

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